12.2 Surface Areas of Prisms and Cylinders

Name _____ Period ____ Date _____

Geometry 12.2 · Student sheet

I can…

- find the surface area of a right prism.
- find the surface area of a right cylinder.
- find a lateral area or a missing dimension from a surface area.

Vocabulary · fill in the missing word

Lateral faces — the faces of a _____ that are not bases

Lateral area — the sum of the _____ of the lateral faces (or the curved surface of a cylinder)

Surface area — the sum of the _____ of all the faces or surfaces of a solid

Net — a _____ pattern that forms a solid when it is folded

Part 1 · Discovery: unfold a box

Cut your box along edges so it lies flat. Measure each face.

Face	Length	Width	Area
Top and bottom (each)			
Front and back (each)			
Left and right (each)			
Total of all six faces			

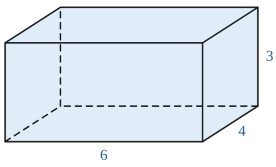
Check: 2 × (area of the bottom) + (perimeter of the bottom) × (height) = _____

Part 2 · Notes and examples (fill in each note, then work the example)

Surface area of a right prism: S = 2B + _____, where B is the area of a base, P is the perimeter of a base, and h is the height.

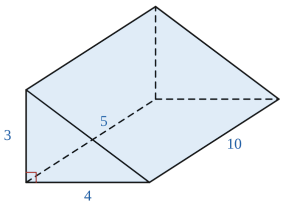
Example 1

Find the surface area of the rectangular prism.



Example 2

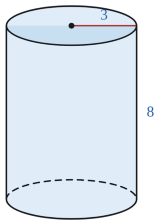
Find the surface area of the triangular prism.



Surface area of a right cylinder: S = 2πr² + _____

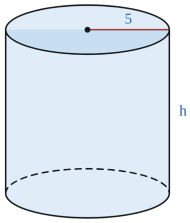
Example 3

Find the surface area of the cylinder. Round to the nearest hundredth.



Example 4

A cylinder has a radius of 5 and a surface area of 150π square units. Find its height.



Lateral area: leave out the _____: L = Ph for a prism, L = 2πrh for a cylinder.

Example 5

A soup can has a radius of 1.5 inches and a height of 4.5 inches. How much paper does its label (the curved side only) need? Round to the nearest hundredth.

Part 3 · Partner practice

Partner 1

Find the surface area of a cube with edges of 5 inches.

Partner 2

A cylinder has a diameter of 10 and a height of 6. Find its surface area. Round to the nearest hundredth.